

SOC Blue Team — Week 2

Threat Detection Lab Report

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Lab Environment: Wazuh 4.x | Manager: SOC01 | Agent: WIN-11 (003) | IP: 192.168.1.8

TASK 1

Brute Force Attack Detection

RULE 60110

Alert Details

Rule ID	60110
Rule Description	Multiple Windows audit logon failures
Rule Level	7 — High
Agent	WIN-11 (ID: 003) IP: 192.168.1.8
Manager	SOC01
Log Source	Windows Security Event Log — EventChannel
Simulation Method	net user Administrator [wrongpassword] x20 via PowerShell loop
Time of Simulation	March 17, 2026 — Lab session
Rule Group	windows, authentication_failures

Short Summary: Is This Brute Force?

YES — This is a brute force attack.

Twenty consecutive failed logon attempts were made against the local Administrator account on host WIN-11, all occurring within approximately 10 seconds. Wazuh fired Rule 60110 — *Multiple Windows audit logon failures* — confirming the behavior exceeded the threshold for what could be considered accidental mistyping.

A legitimate user who forgets their password generates 2 to 3 failed attempts at most before pausing, resetting their credentials, or asking for help. Twenty failures in under 10 seconds from the same source, targeting the same account, is the textbook signature of an automated credential attack — a threat actor systematically cycling through a password list hoping one sticks.

What this attack pattern indicates:

- High-frequency, low-delay attempts — characteristic of scripted/automated tooling, not human input.
- Single target account (Administrator) — attacker is focusing on the highest-privilege local account.
- Same source machine — in a real attack this would be an external IP; here it confirms our simulation worked.
- Rule fired multiple times (firedtimes > 1) — Wazuh's correlation engine confirmed repeated pattern, not a fluke.

SOC Response Action: In a real environment this alert would trigger immediate account lockout review, source IP block at the firewall, and escalation to Tier 2 for investigation of prior successful logins from the same source.

Detection Filter Used

Applied in Wazuh Dashboard → Security Events → Discover:

```
rule.id: 60110
```

```
rule.groups: authentication_failures
```

These filters isolate all brute force correlation events. The time range was set to Last 15 minutes to clearly show the spike in failed logon events generated during the simulation.

Raw Alert — Wazuh Event Detail

Wazuh Rule ID	91816
Rule Description	PowerShell script querying system environment variables
Rule Level	4 — Low-Medium
MITRE ATT&CK; ID	T1082
MITRE Tactic	Discovery
MITRE Technique	System Information Discovery
Event ID	4104 — PowerShell Script Block Logging
Channel	Microsoft-Windows-PowerShell/Operational
Timestamp	Mar 17, 2026 @ 19:45:44.819 UTC
Host	DESKTOP-HDND0QS
Agent	WIN-11 (ID: 003) IP: 192.168.1.8
Manager	SOC01
Process ID	17232
Script Block ID	b2c015fc-6722-47a3-a5d1-890971a63a68
Rule Fired Times	6

Command Captured by Wazuh

```
secedit /export /cfg $env:TEMP\secpol.cfg; Get-Content $env:TEMP\secpol.cfg | Select-String LockoutBadCount; Remove-Item $env:TEMP\secpol.cfg
```

What this command does, broken down:

- **secedit /export /cfg \$env:TEMP\secpol.cfg** — Exports the local security policy to a temporary file. This reveals the machine's account lockout settings, password policy, and audit configuration.
- **Get-Content ... | Select-String LockoutBadCount** — Reads the exported policy file and specifically extracts the lockout threshold — how many failed logins before an account locks. This is reconnaissance.
- **Remove-Item \$env:TEMP\secpol.cfg** — Deletes the evidence file immediately after reading it. This is anti-forensics behavior.

Mini Investigation Report

WHAT HAPPENED

At 19:45:44 UTC on March 17, 2026, PowerShell Script Block Logging (Event ID 4104) captured a three-part command executing on host DESKTOP-HDND0QS. The script exported the local security policy, queried the account lockout threshold (LockoutBadCount), and then immediately deleted the exported file to cover its tracks.

This is not a random or accidental command. The specific extraction of LockoutBadCount is a deliberate reconnaissance step — an attacker who knows the lockout threshold can calibrate their brute force attack to stay just below it, avoiding account lockout while still cycling through passwords. Combined with the immediate file deletion, this is a textbook example of quiet, targeted discovery activity.

WHO DID IT

Host	DESKTOP-HDND0QS
Agent Name	WIN-11 (Wazuh Agent ID: 003)
IP Address	192.168.1.8
Executing User	Not captured in Event ID 4104 — known Script Block Logging limitation when run interactively. In a real investigation this would be correlated with Event ID 4688 (process creation) at the same timestamp to identify the parent process and user context.
Process ID	17232
Timestamp	Mar 17, 2026 @ 19:45:44.819 UTC

RISK LEVEL

MEDIUM-HIGH

Wazuh assigned Rule Level 4, but the true risk is higher when analyzed in context. The script performs reconnaissance (security policy extraction) followed by anti-forensic cleanup (file deletion) — a two-phase pattern associated with pre-attack enumeration.

Risk factors justifying Medium-High classification:

- Maps to MITRE ATT&CK; T1082 (System Information Discovery) — a recognized adversary technique.
- Extracts LockoutBadCount specifically — this data has no legitimate everyday use case; it is primarily valuable for calibrating brute force attacks.
- Immediate file deletion after reading — anti-forensics behavior is a strong indicator of malicious intent.
- Rule fired 6 times — repeated execution suggests automated or scripted activity, not a one-off admin task.
- In isolation the command is non-destructive, but combined with the Task 1 brute force activity on the same agent and same day, this forms a coherent attack chain: reconnaissance followed by credential attack.

COMBINED INCIDENT NARRATIVE: Taken together, Task 1 and Task 2 events on WIN-11 tell a coherent story. At 19:45 the attacker ran a reconnaissance script to extract the account lockout threshold. Armed with that knowledge, a brute force campaign was then launched — timed and calibrated not to trigger lockout. This is exactly the attack chain a real threat actor would follow. A SOC analyst correlating these two alerts on the same host within the same session would immediately escalate this as an active intrusion attempt.

Task Comparison Summary

	Task 1 — Brute Force	Task 2 — PowerShell
Rule ID	60110	91816
Event ID	4625 (Failed Logon)	4104 (Script Block Log)
MITRE	T1110 — Brute Force	T1082 — System Info Discovery
Severity	High	Medium-High (contextual)
Detected By	Wazuh correlation rule	Script Block Logging + Wazuh
Key Finding	20 failed logins in ~10 seconds	Lockout policy recon + cleanup